

VisionInspect AI

Frontend Development Progress Report — Milestone 4

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1. Project Description

VisionInspect AI is an AI-powered quality inspection system that automates defect detection in manufactured products using computer vision and deep learning. Users upload product images and receive defect predictions, heatmaps, and quality decisions.

My role covers Frontend Development, API Integration, Authentication, and Role-Based Access Control (RBAC). Building on the fully functional end-to-end application delivered in Milestone 3, Milestone 4 focused on making the frontend fully mobile responsive and deploying the complete application using Docker and Cloudflare Tunnel.

2. Technologies Used

Category	Technology
Frontend	React.js (Vite), Tailwind CSS, React Router, React Context API
Backend	Flask REST APIs, SQLite Database
Authentication	JWT Authentication
Containerization	Docker, Docker Compose
Web Server	Nginx (serving React build, handling client-side routing)
Model Hosting	Hugging Face Hub (PatchCore model checkpoints)
Deployment / Exposure	Cloudflare Tunnel
Version Control	Git & GitHub

3. Work Completed

Mobile Responsiveness

The entire frontend was made fully responsive across mobile, tablet, and desktop screen sizes using Tailwind CSS breakpoints. Responsive design was systematically applied across all major components:

- Header — collapsible hamburger menu for smaller screens
- Layout — adaptive spacing and structure across screen sizes
- Login & Register pages
- Dashboard (Quality Inspector and Factory Supervisor views)
- Inspection page — upload flow and AI result cards
- History page — search, filters, and record listing
- Result and upload components

This ensures the application is fully usable and visually consistent on any device, which is important for factory-floor and on-the-go usage scenarios.

Deployment (Docker + Cloudflare)

The complete application — frontend, backend, and AI model — was containerized and deployed for team demonstration and evaluation.

- Wrote separate Dockerfiles for the frontend and backend services
- Used Docker Compose to orchestrate both containers together as a single deployable unit
- Configured Nginx within the frontend container to serve the production React build and correctly handle client-side routing
- Externalized all API endpoint URLs into environment variables for portability across environments
- Hosted the PatchCore AI model checkpoints on Hugging Face Hub, with an automated script to download them on container startup
- Set up Cloudflare Tunnel with dual tunnels — one for the frontend (port 3000) and one for the backend (port 5000) — to expose the application publicly
- Validated the full end-to-end flow (login, role-based access, inspection, dashboard, history, analytics, reports) on the deployed instance

4. Deployment Workflow

Stage	Description
1. Build	React frontend built for production; Docker images built for frontend & backend
2. Compose	Docker Compose brings up both containers on a shared network
3. Model Fetch	Backend container auto-downloads PatchCore model weights from Hugging Face on startup
4. Serve	Nginx serves the frontend build; Flask backend exposes REST APIs
5. Tunnel	Cloudflare Tunnel exposes the frontend and backend with public URLs
6. Access	Application accessed via public Cloudflare URLs for demo and review

5. Testing Performed

- Verified responsive layout and navigation across mobile, tablet, and desktop breakpoints
- Tested hamburger menu behaviour and touch interactions on smaller screens
- Validated full application functionality inside Docker containers (auth, inspection, history, analytics, reports)
- Verified AI model auto-download and prediction flow within the containerized backend
- Tested public accessibility and stability of the app through Cloudflare Tunnel URLs
- Confirmed role-based access (Quality Inspector / Factory Supervisor) on the deployed version

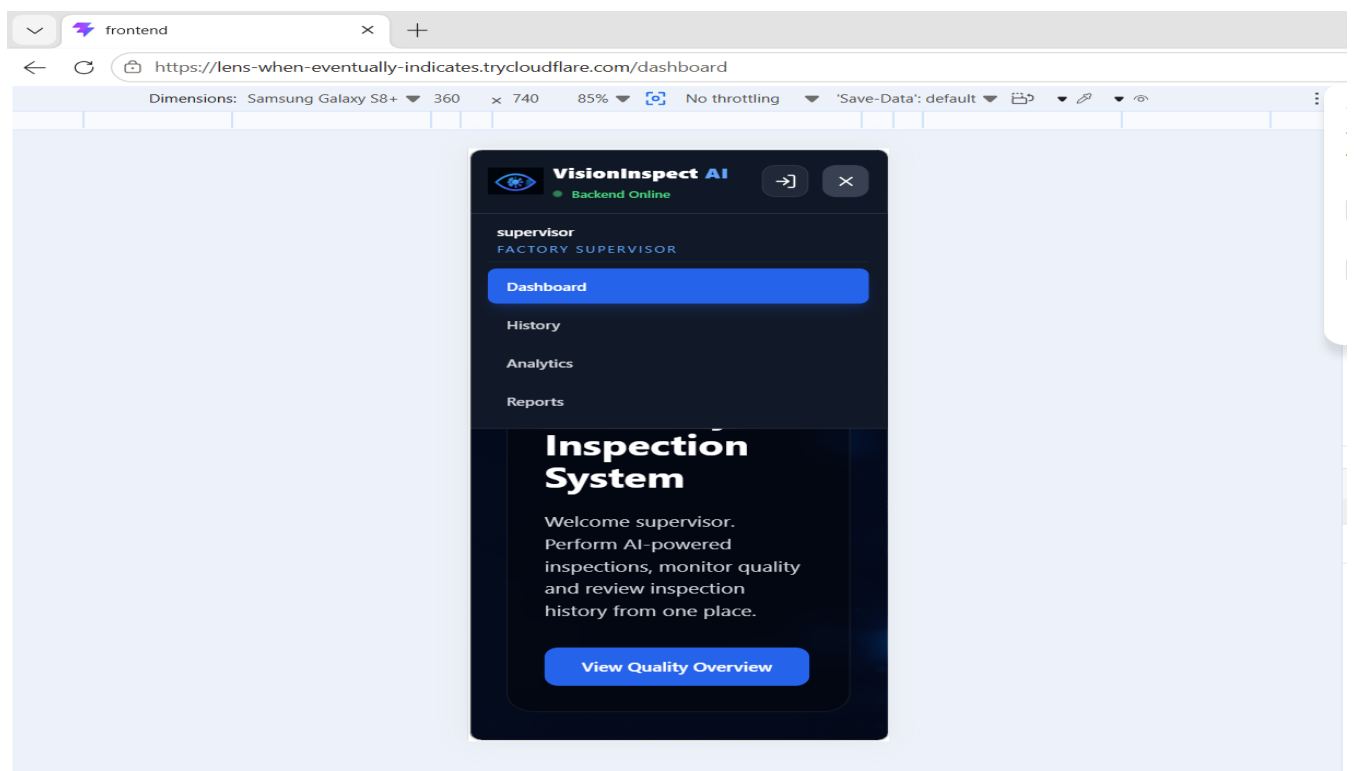
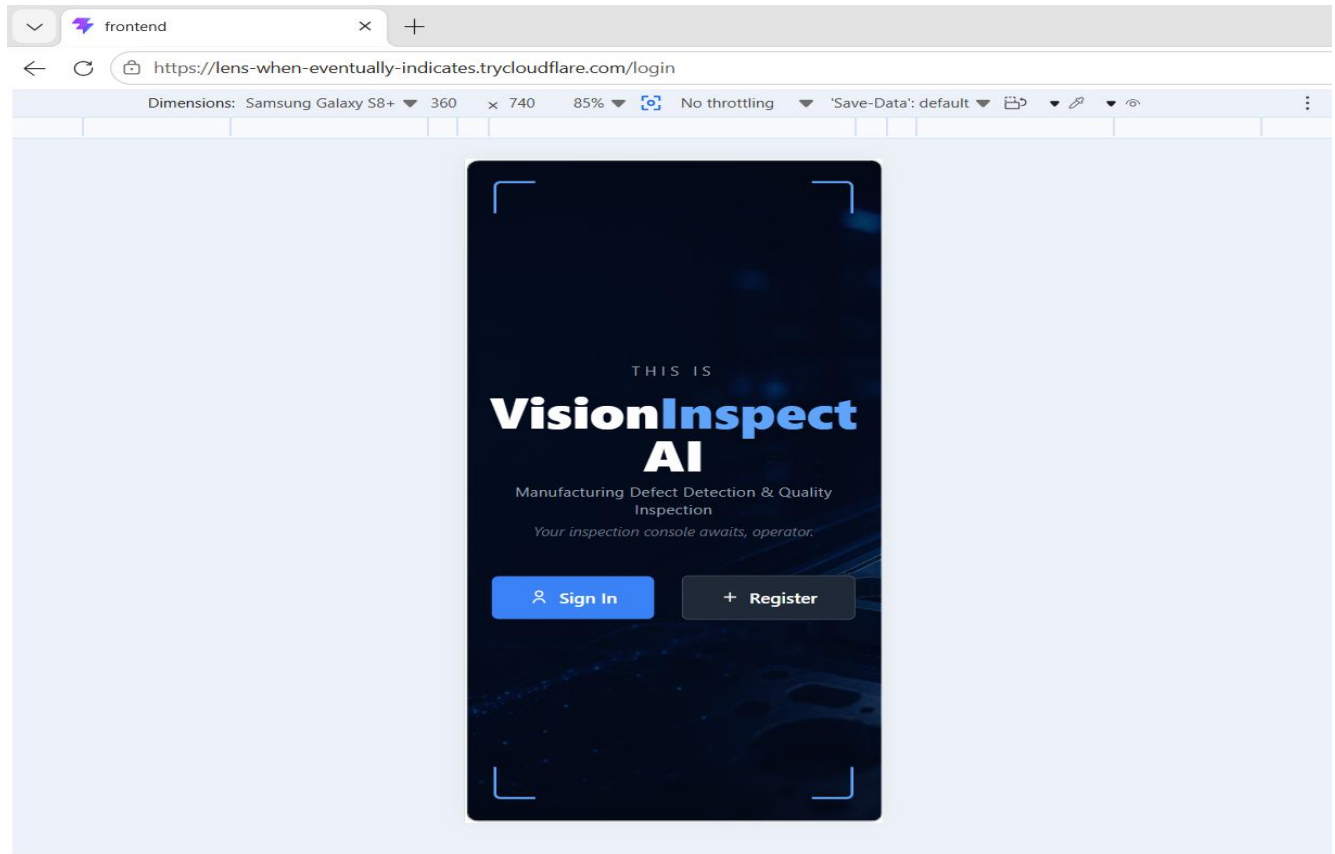
6. Final Outcome

VisionInspect AI is now fully responsive across all device sizes and fully deployed as a containerized application, publicly accessible via Cloudflare Tunnel.

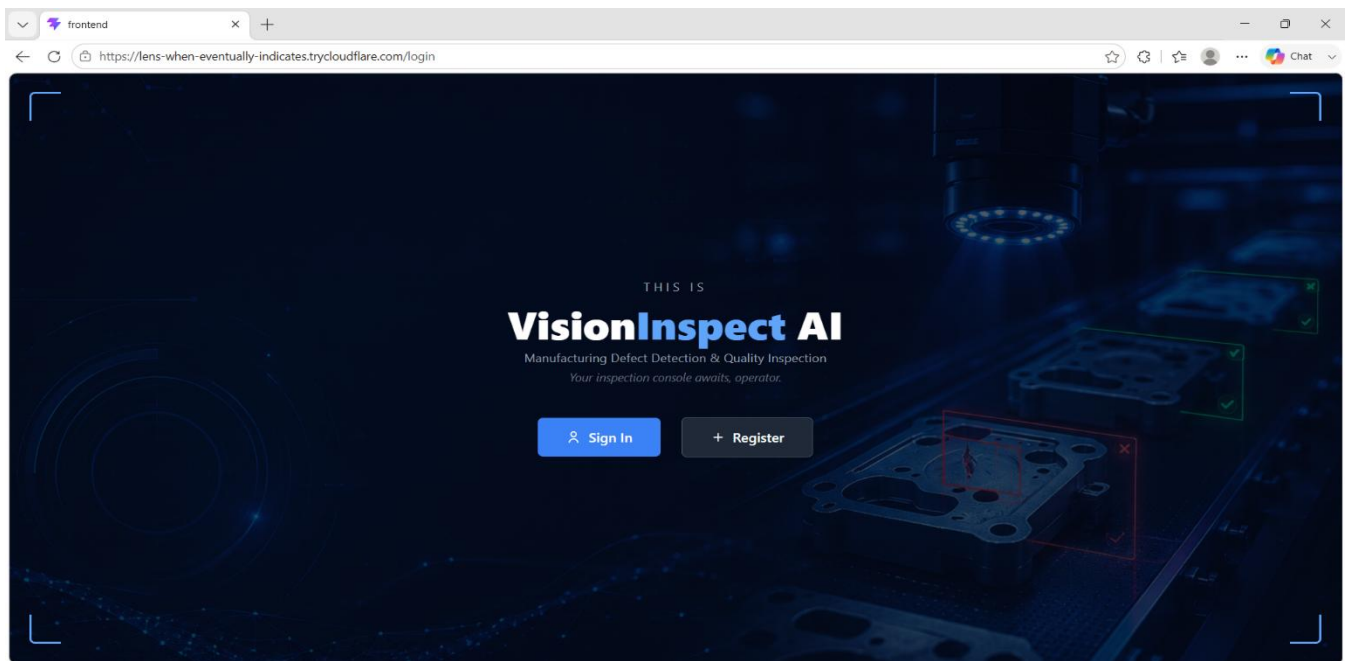
- Fully Mobile-Responsive UI
- Dockerized Frontend & Backend
- AI Model Hosted & Auto-Downloaded via Hugging Face
- Nginx-based Production Serving
- Public Access via Cloudflare Tunnel
- End-to-End Verified Deployment

7. Screenshots

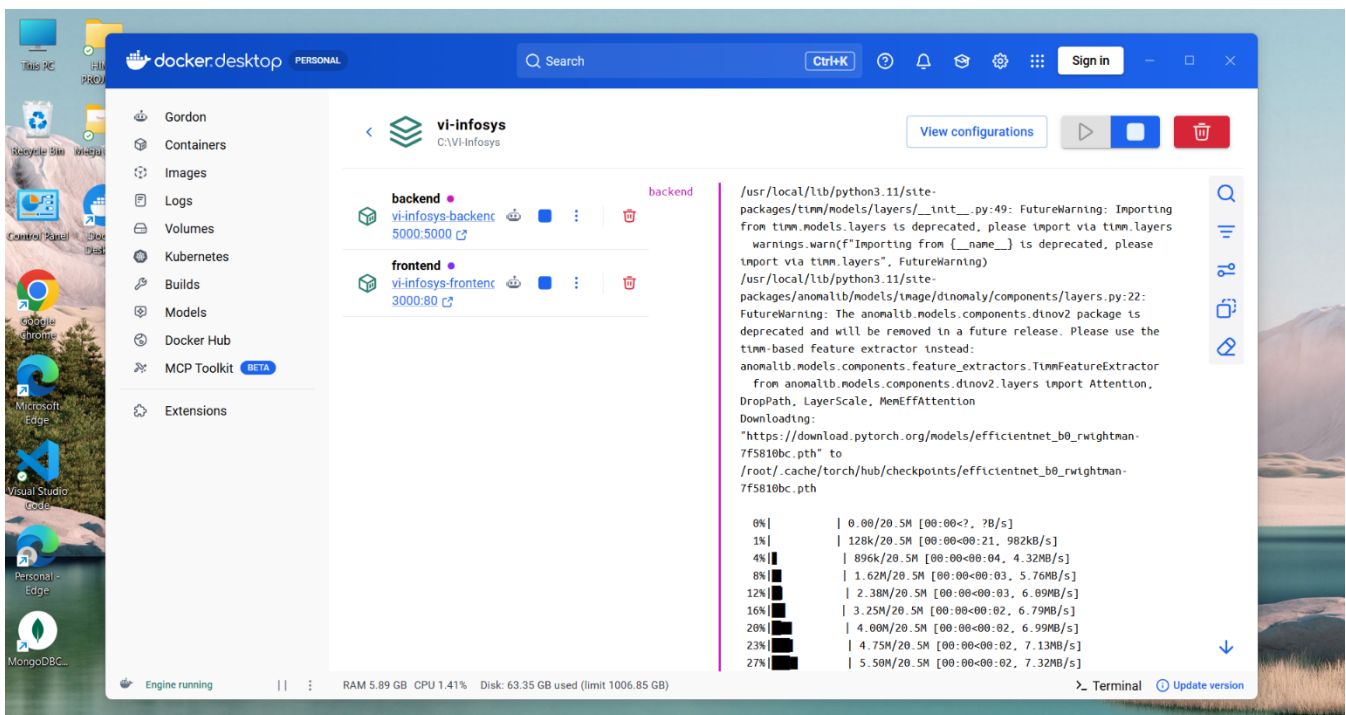
Mobile Responsive



Deployment Link: <https://lens-when-eventually-indicates.trycloudflare.com>



Docker Interface



8. Conclusion

In Milestone 4, the VisionInspect AI frontend was made fully mobile responsive and the complete application was successfully containerized and deployed using Docker and Cloudflare Tunnel. The system is now accessible across devices and available via a public URL, marking the project as fully demo-ready and production-style deployed.